

CART ●

REPORT

State of Spatial Analytics

2026

Foreword

What a difference two years makes. When we published our last State of Spatial Data Science report in 2024, the generative AI revolution was just beginning. Today, its impact is undeniable, reshaping expectations for what's possible in business. It's a pivotal moment for the geospatial community. The conversations we're having now feel different—more urgent, ambitious, and focused on a single, powerful idea: how do we get things done?

The findings in this year's report confirm this shift. We're at an inflection point, moving from creating systems of record to building systems of answers. For years, our community has expertly curated and analyzed data, building the foundational maps that explain our world. But the expectation is changing. The demand now is for instant, accessible insights that don't require a specialist to interpret.

This report reveals both the incredible opportunity and the real challenges of this new era. While AI's potential is clear, its adoption in spatial analytics isn't accelerating. Why? Because tools alone are not enough. The bottleneck isn't a lack of powerful models; it's the persistent friction from fragmented tools, data silos, and inconsistent governance. As this report makes clear, the integration imperative and the move to cloud-native platforms are no longer just trends—they are the non-negotiable foundation for what comes next.

And what comes next is truly exciting. I believe we're transitioning to a new paradigm where the role of the GIS professional evolves from building maps to designing intelligent agents. The road ahead requires us to think differently—to modernize our platforms and, more importantly, our approach. The insights within this report offer a playbook for that journey. I am incredibly excited to see what we, as a community, will build next.



Luis Sanz,

Chief Executive Officer at CARTO

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STATE OF SPATIAL ANALYTICS 2026

Introduction

Executive Summary

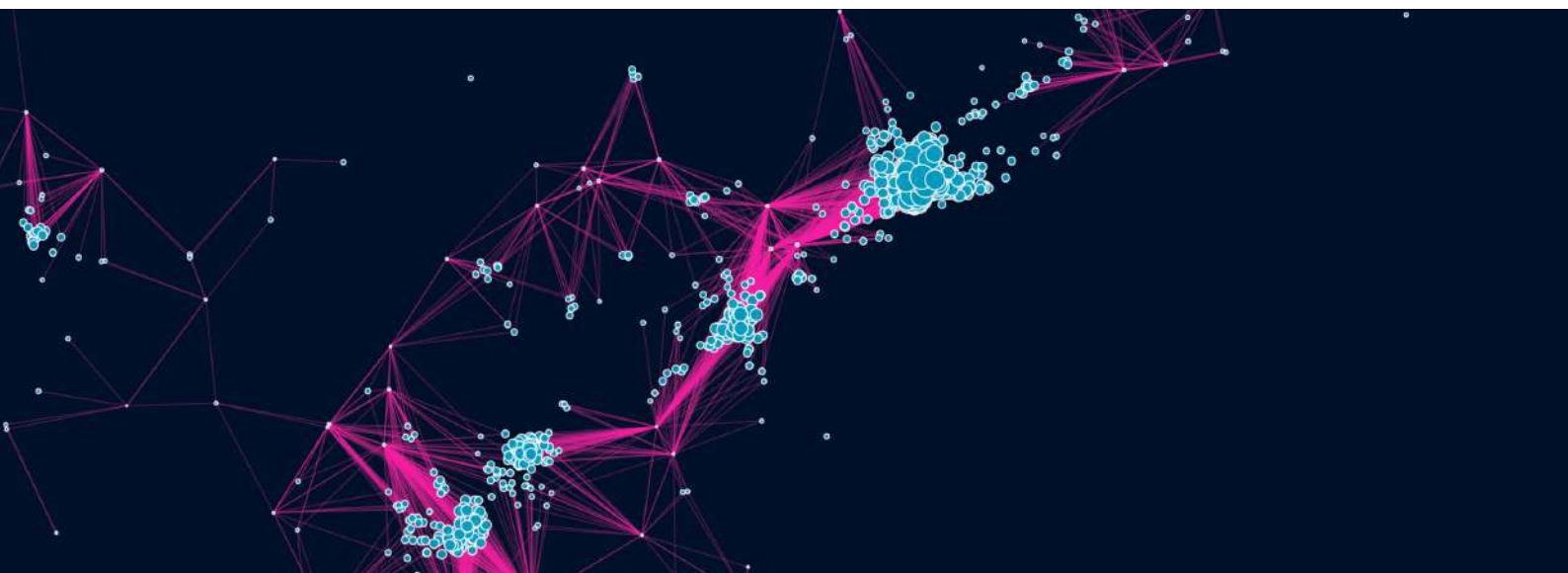
Welcome to the fourth edition of the State of Spatial Analytics report! This resource allows our community to learn about the developments, opportunities, and challenges facing this dynamic and increasingly essential industry.

This year, over 200 professionals responded to the survey, making it one of our most comprehensive views yet. We want to thank everyone who participated and shared their insights, making this report possible.

Key findings from the survey reveal major changes since 2024. A key trend is that the migration to cloud-native platforms has now stabilized, establishing a new enterprise standard. This, combined with a powerful democratization wave, is driving demand for spatial insights to an all-time high.

However, the biggest surprise is that while AI was expected to be transformative, its adoption has stalled. Why has this happened? As the spatial stack becomes more diverse, has the integration imperative become our single biggest bottleneck? And with demand for spatial talent far outstripping supply, how can our community scale expertise to meet these new organization-wide needs?

These trends are the clearest signal for 2026: a strategic shift is required. They show that while new technologies provide great benefits, they also require us to solve new, structural problems. Understanding these dynamics is crucial for navigating the transition from traditional GIS to the intelligent, agent-driven future of our industry.



Spatial Analytics Evolution Timeline

2026 Key trends

The key trends from the 2026 State of Spatial Analytics Survey are:



AI as the Force Multiplier

AI is a force multiplier, not a replacement



The Integration Imperative

Integration emerges as the biggest bottleneck



The Enduring Scarcity of Spatial Talent

Spatial talent is as scarce as ever



The Democratization Wave

Democratization is the new normal



Governance Becomes Mission-Critical

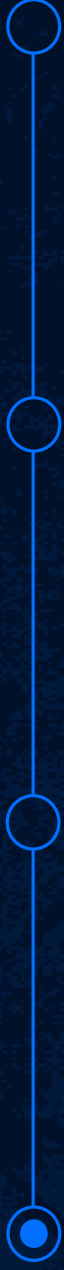
Data governance becomes critical, fueled by AI demands



The Cloud-native Prerequisite

Cloud-native, centralized, and accessible data is a prerequisite

Looking back at previous editions of this report.

A vertical timeline graphic on the left side of the page. It consists of a vertical line with four circles. The top three circles are empty, and the bottom circle is filled with a blue dot. The years 2020, 2022, 2024, and 2026 are listed next to these circles respectively.

2020

As GIS extends well beyond its traditional community, the value and application of Spatial Data Science is now seen in an all-new light.

2022

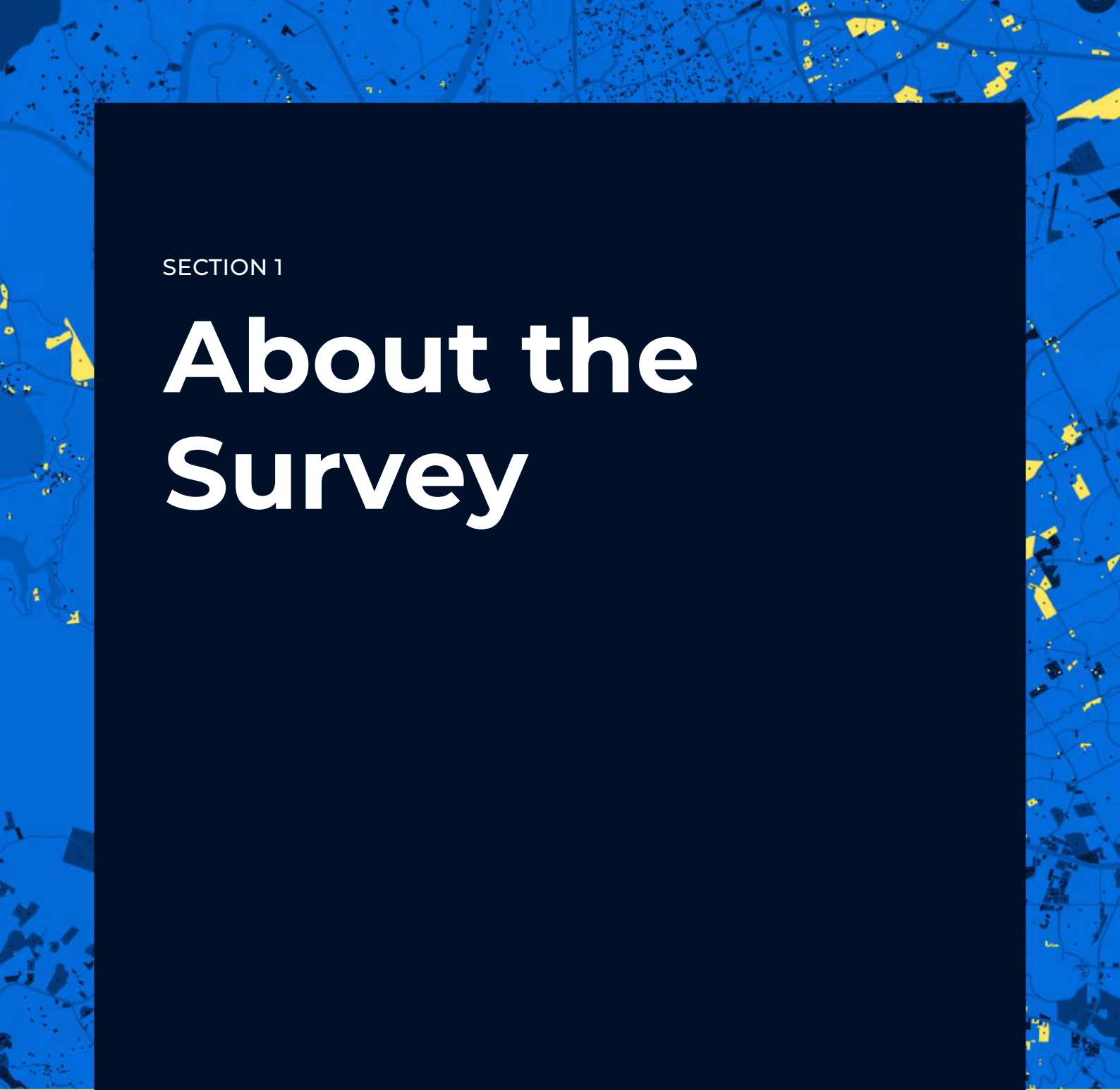
Spatial Data Science is no longer a niche specialty within data science teams; its use case is quickly becoming mainstream.

2024

Spatial Data Science's expanded reach is fueling new opportunities and innovations around AI, skills development, and interoperability.

2026

Spatial Data Science and more broadly Spatial Analytics, is at an inflection point, setting the stage for an agentic, cloud-native future while grappling with major challenges in integration and data governance.

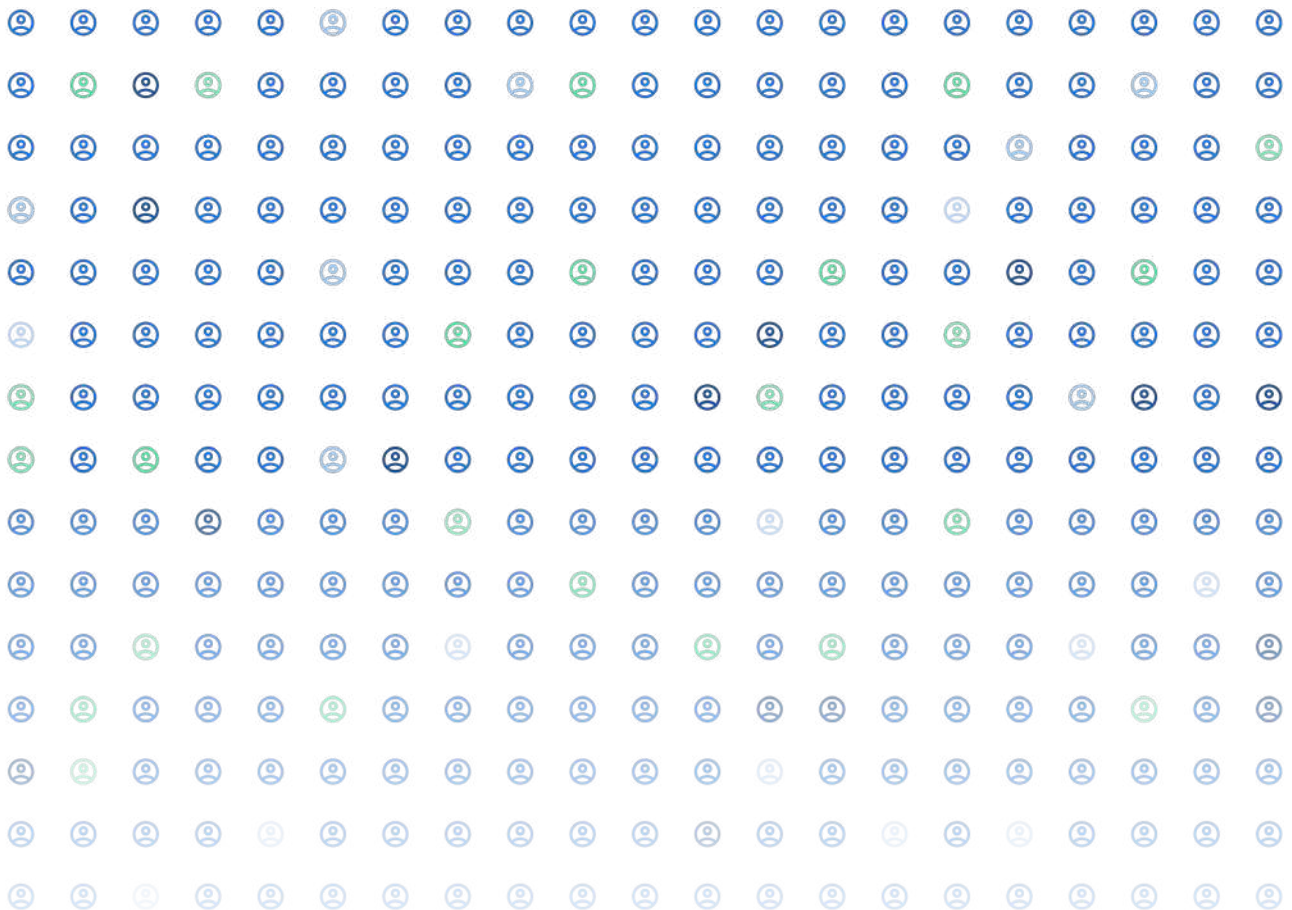


SECTION 1

About the Survey

Who participated in the survey?

This year's survey is one of our most comprehensive yet. Over 200 professionals shared their perspectives on the evolving state of Spatial Analytics.



Industry participation closely mirrors trends from previous editions, with the highest number of responses coming from:

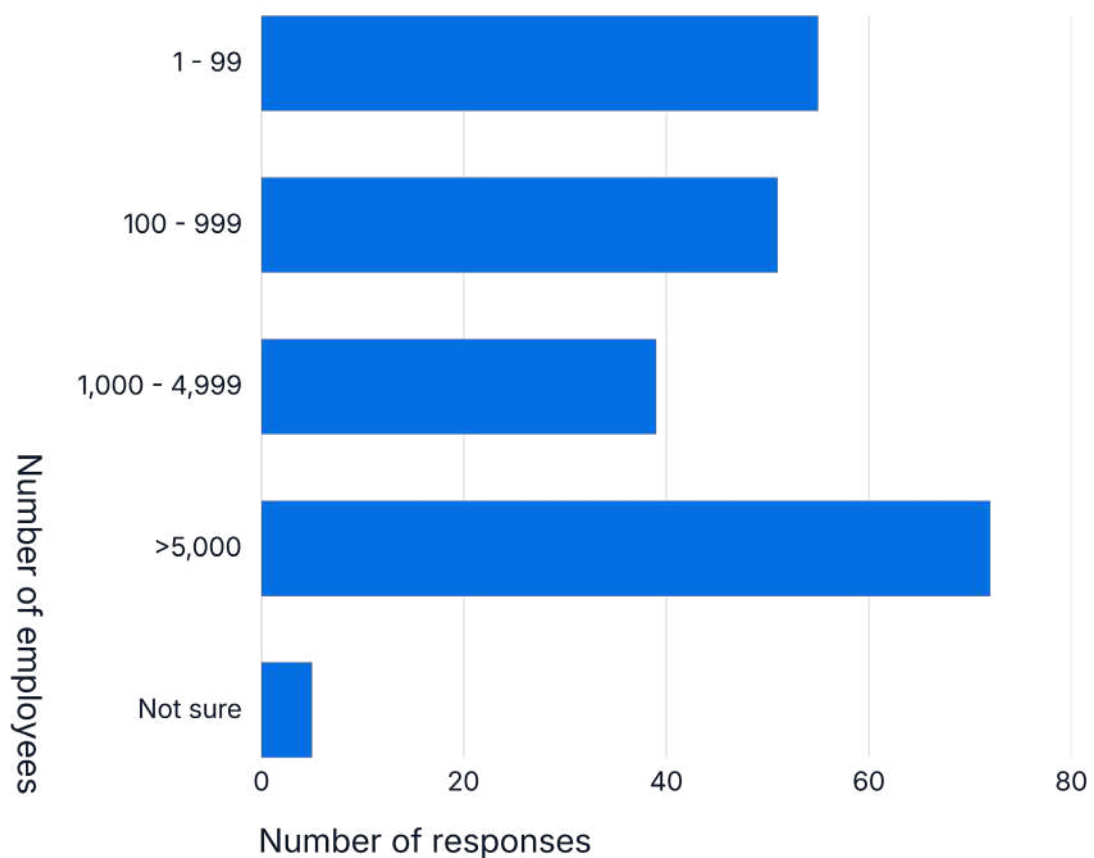
- Software & Technology - **17.6%**
- Cities & Government - **9.7%**
- Natural Resources & Environment - **8.4%**
- Consulting - **7.6%**

Explore the full range of industries covered on the following page.

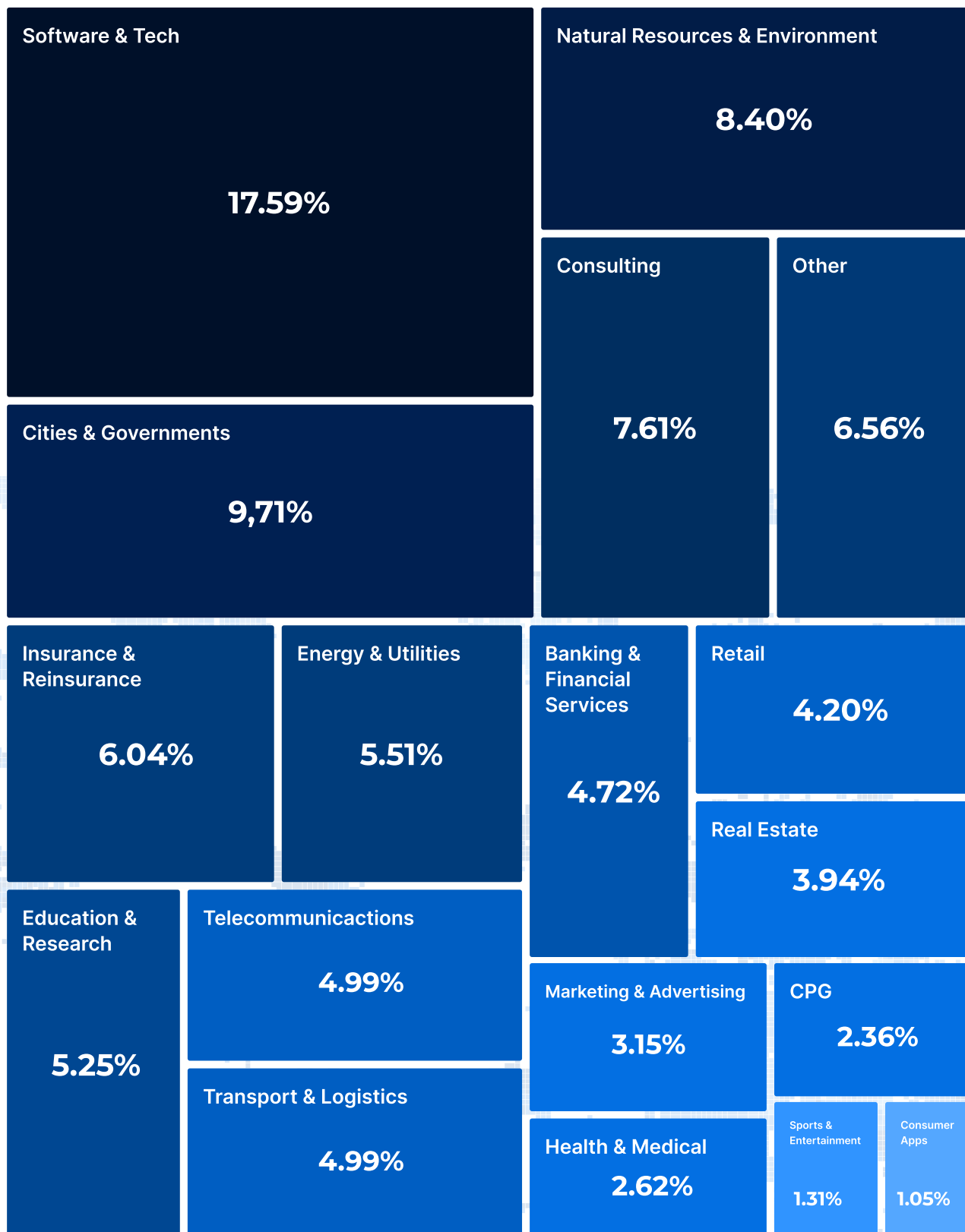
While responses came from companies of all sizes, the majority were from large enterprises, reinforcing the growing strategic importance of Spatial Analytics within complex, global organizations.

Together, this mix of roles, industries, and enterprise scales creates one of the clearest views yet of how Spatial Analytics is practiced today - and where it is heading.

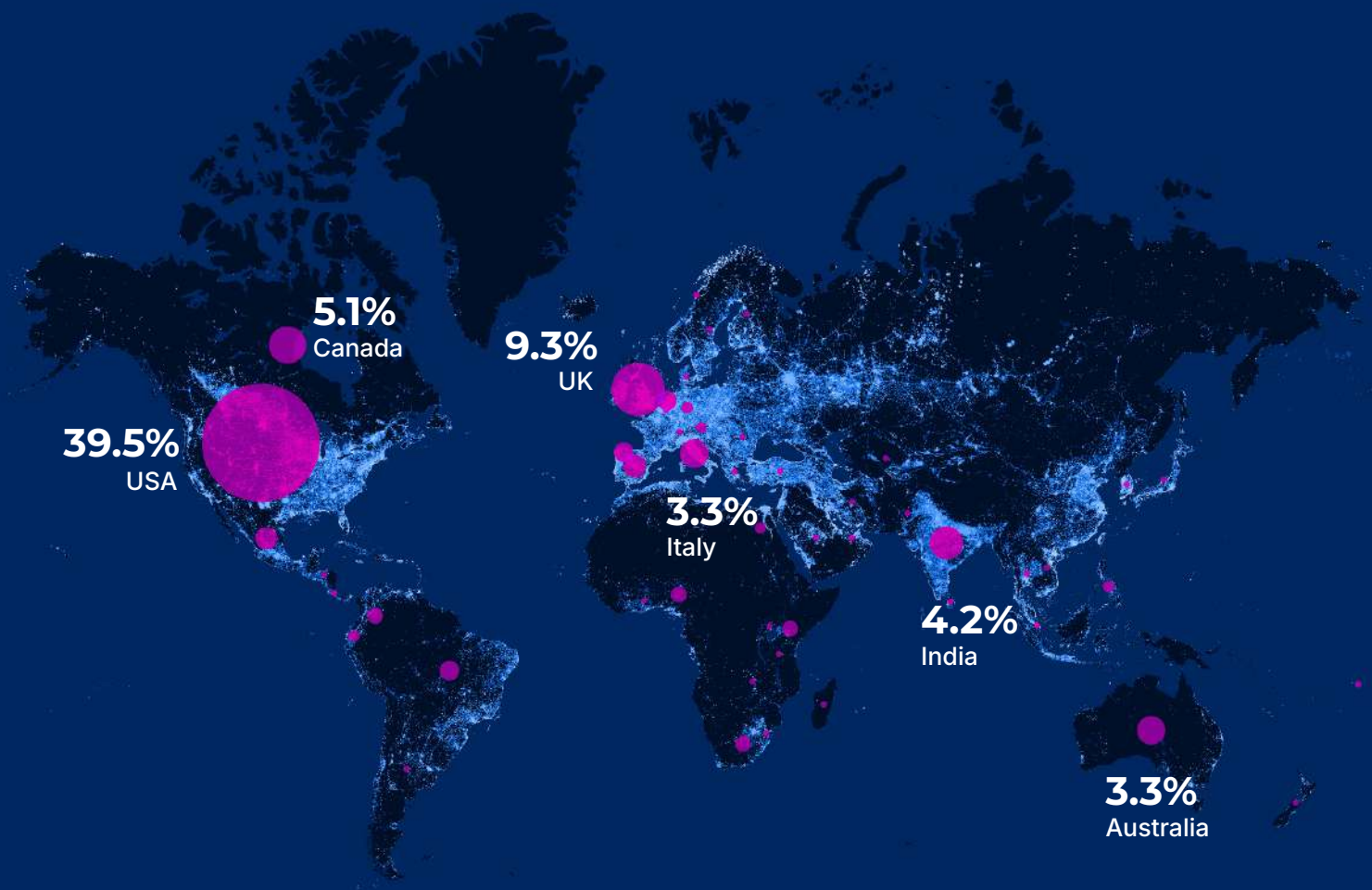
How many people work at your organization?



Which industry best describes the one you work in?



In which country are you based?



SECTION 2

Spatial Analytics in 2026



2026: AI's breakthrough moment in spatial analytics?

AI moves from experimentation to essential.

The findings from this year's survey confirm that AI is now an established component of spatial analytics. But despite two years of significant advances in AI technology, the foundational adoption rate remains steady, indicating that its integration is consistent but not accelerating at a breakneck pace.

Most of this is bottom-up - rather than organizations embedding AI into their organizational processes, adoption is coming from the teams actually doing the work - using copilots, LLMs & AI Agents to boost their productivity. While the broader enterprise world reports near-universal AI adoption, GIS remains defined by an entrenched divide between early and late adopters.

The blockers are structural: **rising concerns about data security and privacy**, uneven governance maturity, unclear regulation, and skill gaps. Organizations increasingly understand AI's potential, but they're not yet confident they can deploy it safely, responsibly, or at scale. The challenge ahead is not discovering what AI can do for spatial analytics - it's building the conditions that allow it to flourish.



The AI divide

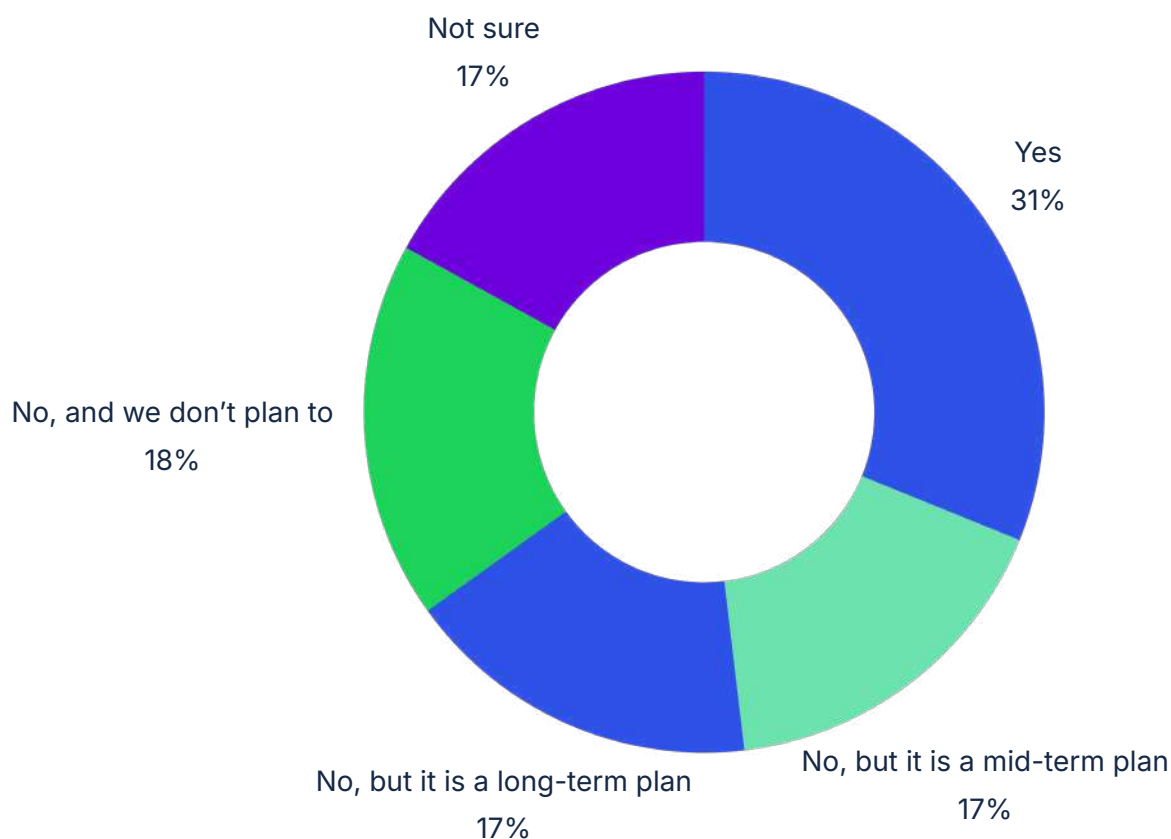
While the rapid adoption anticipated by the hype hasn't fully materialized, our survey indicates a steady and encouraging presence of AI in geospatial, maintaining its traction over the past two years.

Just 31% of organizations have invested in AI tools - almost identical to in 2024 - while those treating AI as a mid or long term plan have only risen modestly from 26% to 34%. Even more noticeably, the share with no plans at all has grown from 15.4% to 17.9%.

This suggests that the long-standing gap between early and late adopters in GIS is becoming even more entrenched. And the contrast with the wider enterprise world is striking: 88% of companies globally now use AI in some form (source: [McKinsey](#)).

So why is spatial analytics moving so slowly? In this chapter, we dig into what's driving this.

Has your organization invested in AI tools?

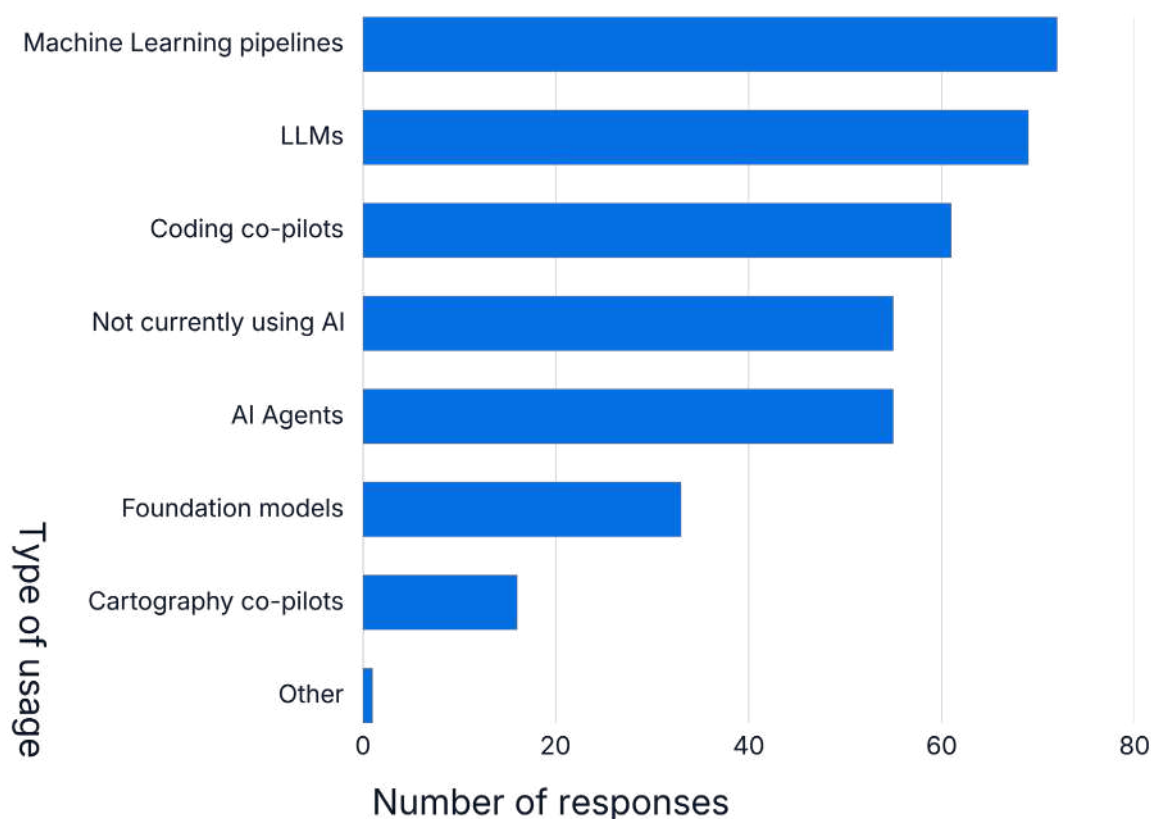


How AI is being used today

AI is entering Spatial Analytics from the **bottom up** rather than through sweeping organizational strategy. The most common pattern is individual use of productivity tools (45%), showing that AI is being adopted organically by practitioners before it's being formalized by leadership. Only 18.3% say AI is embedded into their organization's processes - clear evidence that the industry is still experimenting rather than standardizing.

When we look at what teams are actually using AI for, a similar picture emerges. Machine learning, LLMs and coding copilots lead the way - practical, tactical applications that help analysts move faster on everyday tasks. Meanwhile, more advanced or enterprise-grade capabilities such as AI agents and foundation models are still in their early days.

Which practical applications is your organization using AI for in spatial analysis?

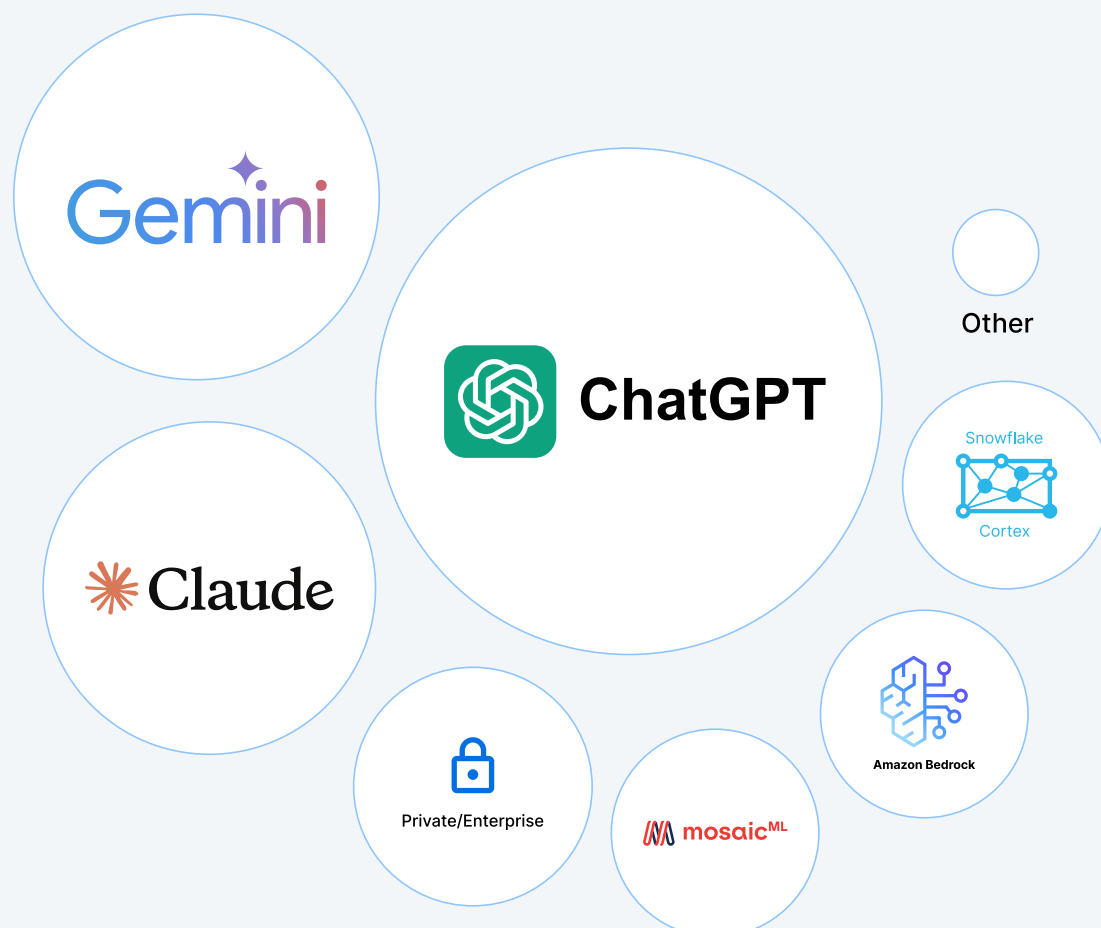


Tools, platforms & real-world adoption

In terms of tools, the ecosystem is dominated by a few major players. **ChatGPT leads at 56%**, followed by **Google's Gemini / Vertex AI (30%)** and **Anthropic's Claude (26.7%)**. Enterprise-grade platforms - like **Databricks Mosaic (11.1%)**, **Snowflake Cortex (7.2%)**, and **Bedrock (9.4%)** - make up a smaller but growing share. This suggests that spatial organizations are still in a learning and prototyping phase rather than fully integrating AI into enterprise pipelines.

While some teams are embracing AI fully, most organizations are still in the early-to-middle stages of adoption. When asked how many people in their organization use AI in their day-to-day work, about one-third estimated **roughly half of the organization**, while only 16% said "almost everyone" is using AI. This shows that, although AI is becoming more common, it has yet to reach widespread, organization-wide use.

Which LLMs and related AI tools is your organization using to support spatial analysis?



"It all ties into agents, taking on and shouldering the busy work to enable you to focus on the things that add value and to the bottom line of the business."



Jon Poole

Enterprise Account Executive at Snowflake

Barriers to AI adoption

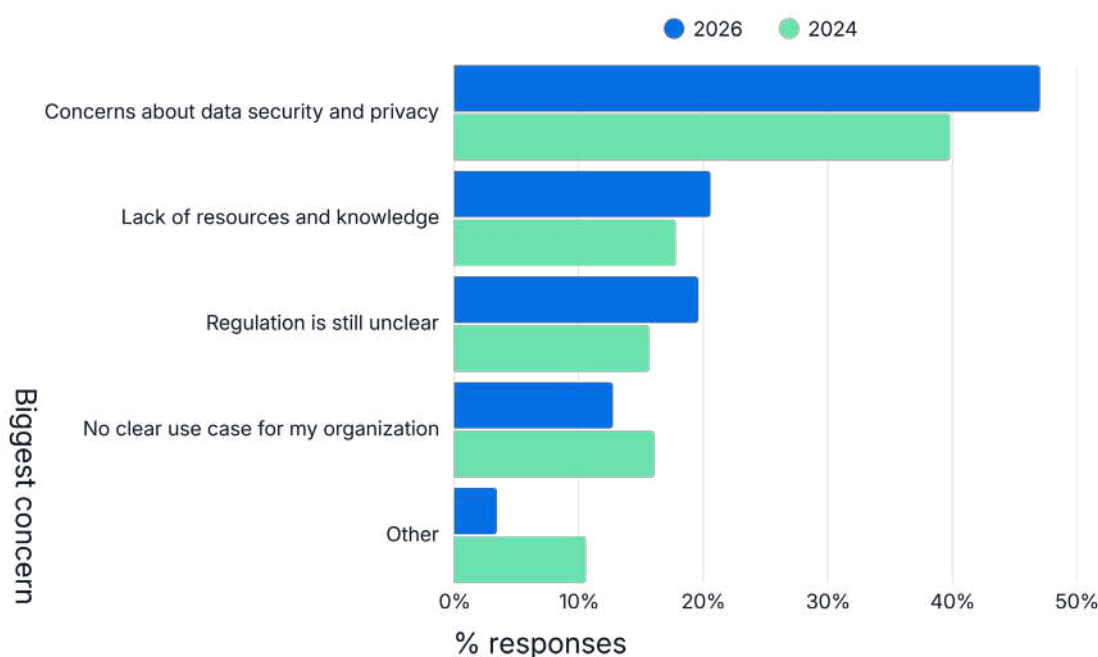
While AI adoption in GIS is progressing at a steady pace since our last survey, it's encouraging to see that organizations are actively addressing the evolving concerns that accompany this growth.

The biggest jump is in security and privacy fears, rising from **39.8% in 2024** to **47% in 2026** - making it by far the **biggest blocker to spatial AI adoption**. With many geospatial datasets tied to infrastructure, people, and assets, this heightened caution is understandable.

Other barriers have crept up too. Lack of skills and resources increased from 17.8% to 20.6%, and regulatory uncertainty rose from 15.7% to 19.6%. Meanwhile, fewer respondents say they lack a clear use case (down to 12.6%). A small number also raised concerns about the environmental impact of AI, adding a new dimension to the hesitation.

This all suggests that organizations are becoming clearer on what AI could do for them - they're just **not confident they can deploy it safely**, responsibly, or in compliance with emerging rules.

What is the biggest concern for your organization when it comes to AI?



"We understand that safeguarding data is paramount. Even as customers embrace powerful large language models, they expect unwavering confidence that data guardrails remain resilient and impenetrable. ***Upholding the highest standards of data integrity is simply non-negotiable.***"



Michael Johns

Geospatial Specialist Leader at Databricks

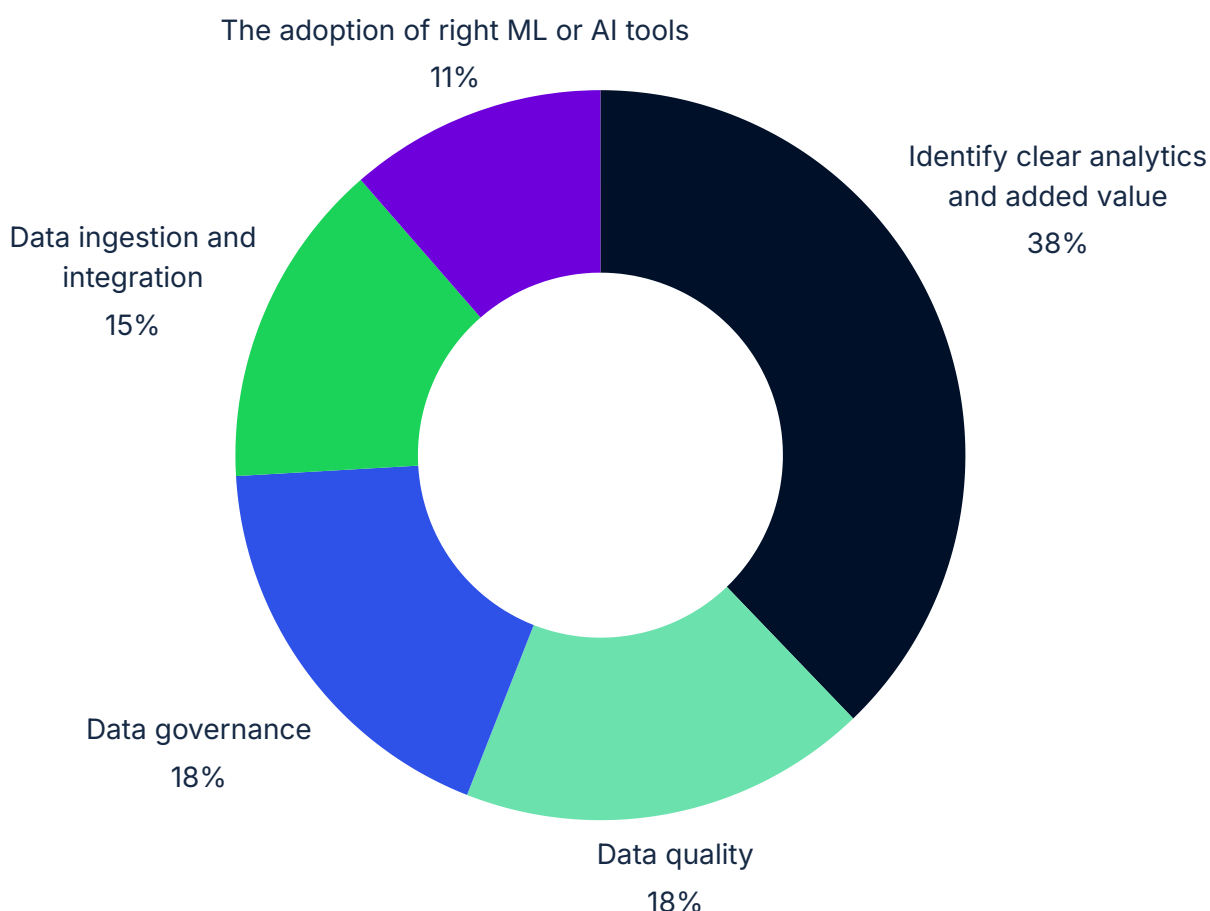


Building the right AI strategy

A successful AI strategy doesn't begin with choosing models - it begins with defining impact. The experts we surveyed overwhelmingly say the first step to building a successful AI strategy is identifying where AI can deliver real analytical value (38%). That's the strategic gap: organizations aren't short on tools; they're **short on clarity about where AI moves the needle**.

What comes next is the foundation everything else depends on: data quality, governance and integration. Without trusted, well-governed spatial data, AI simply can't be deployed responsibly or at scale. It's telling that "choosing the right tool" was the least important factor (11%). The takeaway? Technology isn't the bottleneck - getting **your data** and your **priorities** right is.

In your opinion, how important are the following considerations when developing an AI strategy?



Your 2026 spatial playbook:

AI Adoption

With the spatial analytics industry being overall slow on capitalizing on the benefits of AI, here's how you can beat the competition!

✓ **Step 1 • Start with one high-value workflow.**

Don't start with tools; start with impact. Identify one repeatable spatial workflow that slows your teams down today - something repeatable, cross-functional, and business-critical that is causing bottlenecks across your organization.

✓ **Step 2 • Make the data behind it water-tight**

Strengthen the basics: clean data, governance, integration, lineage. If the data feeding AI isn't trusted, neither will the results be. Governance is the real accelerator.

✓ **Step 3 • Bring AI to your data, not the other way around**

Run AI inside your cloud data warehouse or governed environment. Connect LLMs, copilots, or AI agents directly to your spatial data - don't move your data to the model. This reduces risk and increases adoption instantly.

✓ **Step 4 • Empower everyone.**

Upskill everyone. Set up your analysts to build, monitor, and refine AI-driven spatial workflows - with everyone else asking natural-language questions and getting instant insights. When you start breaking those GIS silos and bottlenecks, adoption becomes organic.

✓ **Step 5 • Measure, learn & scale**

Start small. Prove value fast. Then expand. Track where AI saves time, reduces manual work, accelerates analysis, or improves decisions - and scale only the workflows that demonstrate real impact.

The bottom line: AI doesn't transform geospatial work through tools alone. It transforms organizations by strengthening data foundations, redesigning workflows, and **enabling people**. Start focused, build safely, and scale what works. The organizations that do this now will define the next era of spatial analytics.



02

Why cloud-native has become non-negotiable

Spatial Analytics moves to where your data lives, not the other way around.

Cloud-native spatial analysis has moved from a trend to a baseline expectation. With **68.5% of organizations now running GIS in the cloud**, the migration has stabilized after years of rapid growth. Cloud-first isn't just about technology - it's about performance, governance, scalability, and collaboration. Organizations still tied to legacy tools or siloed databases risk falling behind in speed, analytical reach, and enterprise impact.

Adopting the cloud fundamentally changes how spatial data is managed and used. True cloud-native analytics means processing, visualizing, and analyzing data where it lives - inside your warehouse - without copying or syncing. This approach unlocks elastic compute, tighter governance, centralization of business assets, and enterprise-ready systems that embed spatial analytics directly into business decision-making.

Spatial Analytics on the cloud

This year's results reveal a clear but stabilizing trend: 68.5% of respondents report running their spatial analysis in the cloud, a change of just **one percentage point** since 2024. After years of rapid migration - including a 14-point jump between 2022 and 2024 - it's clear that cloud-native GIS has become the norm.

Though cloud adoption may be plateauing, the message is unmistakable: spatial analytics has matured into a **cloud-native standard**. Organizations not running GIS in the cloud risk falling behind in performance, scalability, and collaboration. Cloud-first isn't just a trend - it's a pivotal baseline organizations need for modern, enterprise-ready spatial analytics.

Do you use a cloud data warehouse, data lake or lakehouse for spatial data?

68.5%
Yes

31.5%
No

"Once we got everything loaded into the cloud other use cases started presenting themselves, and some that weren't even map-forward. We really saw the power of the entire ecosystem and started to leverage off of that."

GIS Director at a leading global advertising company



What does cloud-native mean in 2026?

Cloud-native spatial analytics means running your geospatial processing, visualization, and data science directly inside your cloud data warehouse. No syncing, no copies, no hidden databases. Compute is pushed down to the warehouse, security and governance stay intact, and teams tap into elastic cloud performance without moving a single row of data.

The payoff is huge: faster analysis, cleaner governance, and maps that scale to millions - or billions - of points and polygons. Here are some of the benefits found in the survey:

Organizations running spatial analysis on the cloud are...

1.7x

more likely to have enterprise-grade systems in place, and have spatial analytics as a core function in their organization.

29%

less likely to cite data discovery, sourcing & cleaning as their top challenge when working with spatial data.

19%

less likely to cite limited computational resources as their top bottleneck.

45%

less likely to struggle with collaboration across teams.

24%

more likely to see demand for analytics rise from non-technical colleagues.

59%

less likely to report software selection as "very difficult."

4x

more likely to have AI embedded into their organizational processes, whilst organizations not on the cloud are 68% more likely to see individuals using AI as ad-hoc productivity tools.

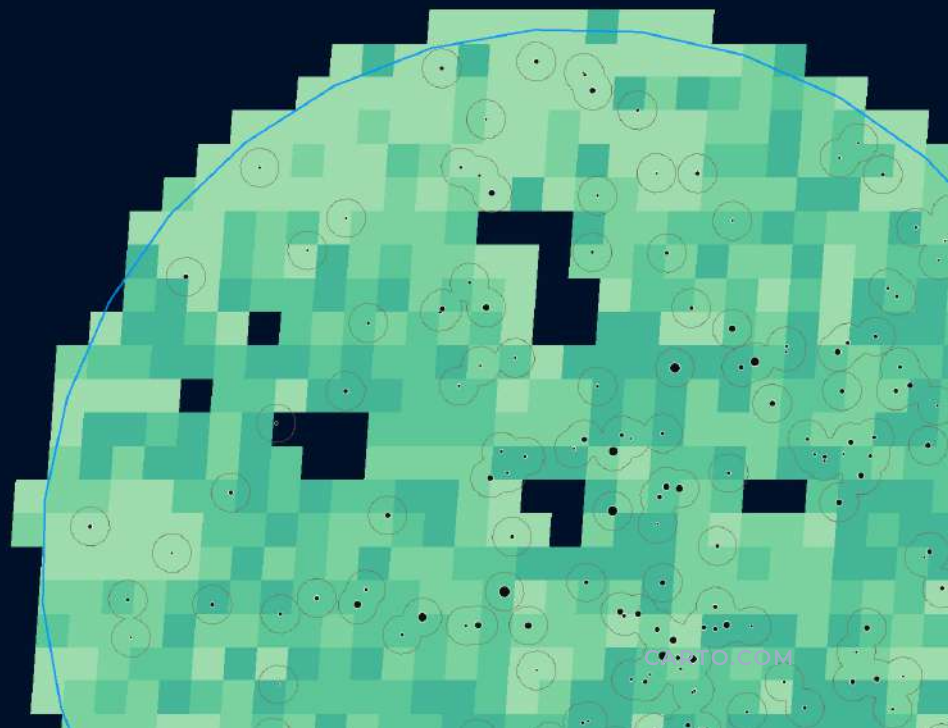


"With cloud-native spatial analytics in 2026, integrating AI capabilities becomes straightforward. Because analysis happens directly in the cloud data warehouse or lakehouse, organizations can more easily apply AI to geospatial data, or expose workflows to AI agents, helping **teams gain deeper insights and enabling non-experts to unlock value from location data**, all while maintaining security and governance."



Siva Ravada

Vice President, Software Development, Oracle

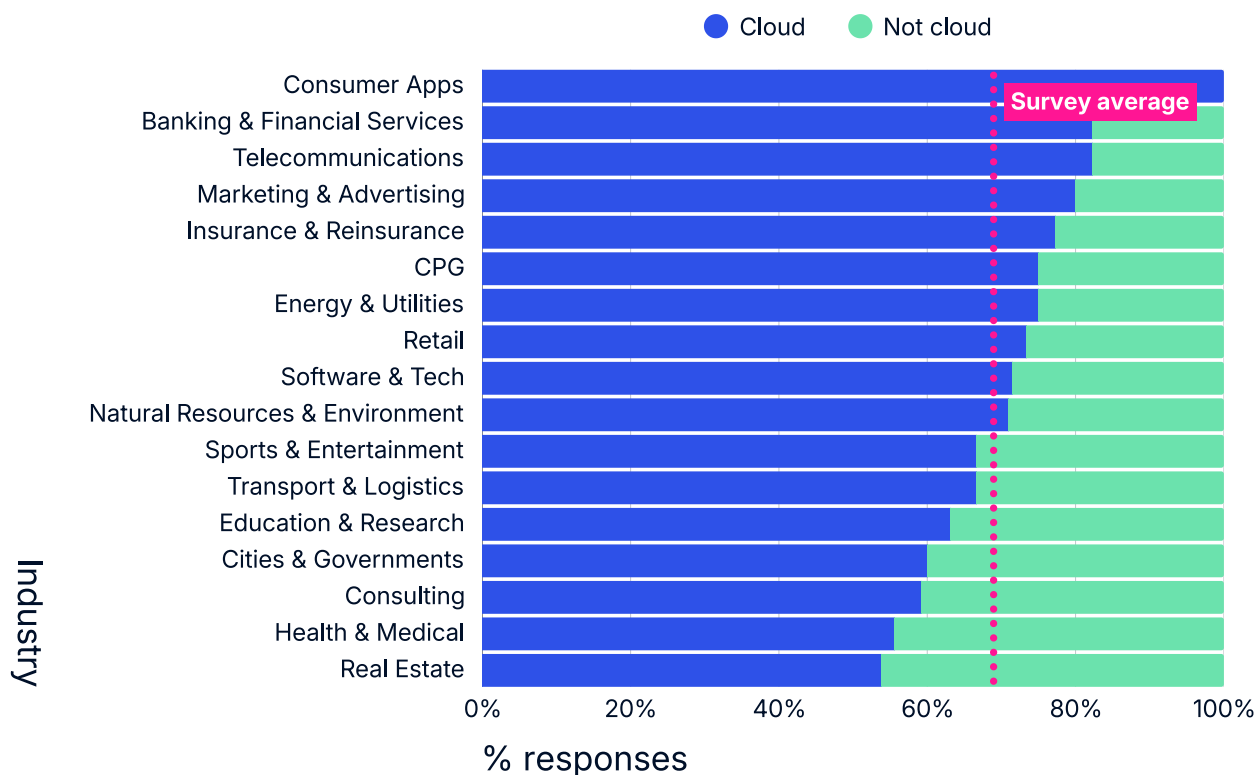


How are organizations leveraging the cloud?

Cloud-native spatial analytics is now the default across most major industries - but the **gap between leaders and late adopters is widening**. At the top, Consumer Apps hit 100% adoption (although it should be noted the small number of respondents in this industry). Meanwhile, Banking, Telecoms, Marketing, and Insurance - sectors that rely on massive, fast-moving datasets and tight governance requirements - all exceed 77% cloud adoption. These industries have fully embraced the warehouse-as-platform model because it delivers what they need most: performance, security, and real-time scale.

At the other end of the spectrum, cloud adoption is lowest in Health & Medical (56%), Real Estate (54%), and Consulting (59%). This could reflect industries with stricter compliance, fragmented data landscapes, or legacy vendor dependencies and lock-in contracts. The takeaway? Cloud-native is becoming a competitive advantage - especially among sectors where agility, data integration, and analytical scale directly drive growth.

Are you running spatial analytics on the cloud?



Your 2026 spatial playbook:

Taking full advantage of the cloud

So you're considering or have made significant investments in a cloud ecosystem - congratulations! Now, here's how you can unlock its full potential for spatial analytics:

✔ **Step 1 • Eliminate GIS and data silos**

Analyze your spatial data where it already lives. Seek platforms that let you run advanced spatial analytics directly in your warehouse, so you don't let your spatial data fall into a silo.

✔ **Step 2 • Leave static analysis behind**

Move beyond static maps, dashboards, and reports. Take full advantage of the cloud with real-time analytics, predictive spatial models, and scalable dynamic visualizations for faster, smarter decision-making.

✔ **Step 3 • Democratize Location Intelligence**

The cloud is designed for collaboration - so make sure your spatial analysis is too. Empower teams of all skill levels to ask questions, explore patterns, and get answers instantly - making spatial intelligence a shared capability.

✔ **Step 4 • Integrate spatial analytics into your cloud ecosystem**

Don't let spatial data live in isolation. Connect your cloud-native GIS to LLMs, AI agents, operational workflows, APIs, and cloud pipelines so insights automatically feed into business processes.

✔ **Step 5 • Innovate and test with confidence**

Push the boundaries of your spatial analysis with AI agents, advanced modeling, or try new analytics techniques while keeping governance, security, and compliance intact. Innovate at speed without compromise.

The bottom line: Cloud-native GIS tooling is the non-negotiable foundation for modern spatial intelligence. Organizations that take full advantage of this in 2026 will unlock speed, scale, collaboration, and organizational insights like never before.



03

The integration imperative

Your spatial tools are only as powerful as the ecosystem they connect to.

The 2026 survey reveals a spatial ecosystem that's more diverse - and more fragmented - than ever. Most organizations now rely on a constellation of GIS tools, BI platforms, open-source libraries, and data engineering frameworks to deliver spatial insights.

Yet this breadth comes with a cost: **integration** is now the single biggest bottleneck in the spatial workflow. Nearly 30% of respondents cite data access and integration delays as their primary blocker, far outpacing issues like slow processing or collaboration. With teams juggling anywhere from three to eight tools on average, the risk of duplicated data, inconsistent logic, and slow iteration grows exponentially.

And survey respondents are feeling that strain. Themes like license complexity, scattered tooling, difficulty collaborating, and fear of vendor lock-in all point to the same underlying problem: spatial analysis only works at scale when **tools work together**.

Integration is the biggest bottleneck organizations face

Integration remains the top bottleneck for organizations doing spatial analysis. Nearly **30% of respondents** report that data access and integration delays hold their teams back, ahead of challenges like data cleaning, collaboration, or slow processing. The modern spatial stack is powerful, but only if tools can work seamlessly together - otherwise, teams spend more time wrangling data than generating insights for strategy and operations.

Complexity around licensing, extensions, and fragmented platforms compounds the problem. Several survey respondents noted that sharing and collaborating across teams, harmonizing multiple tools, and balancing cost versus flexibility are ongoing challenges.

What do you find particularly difficult about finding the right software for Spatial Analysis?

"Being able to share and collaborate with others - and ease of set up"

Geospatial Consultant working in Geospatial Consulting, USA

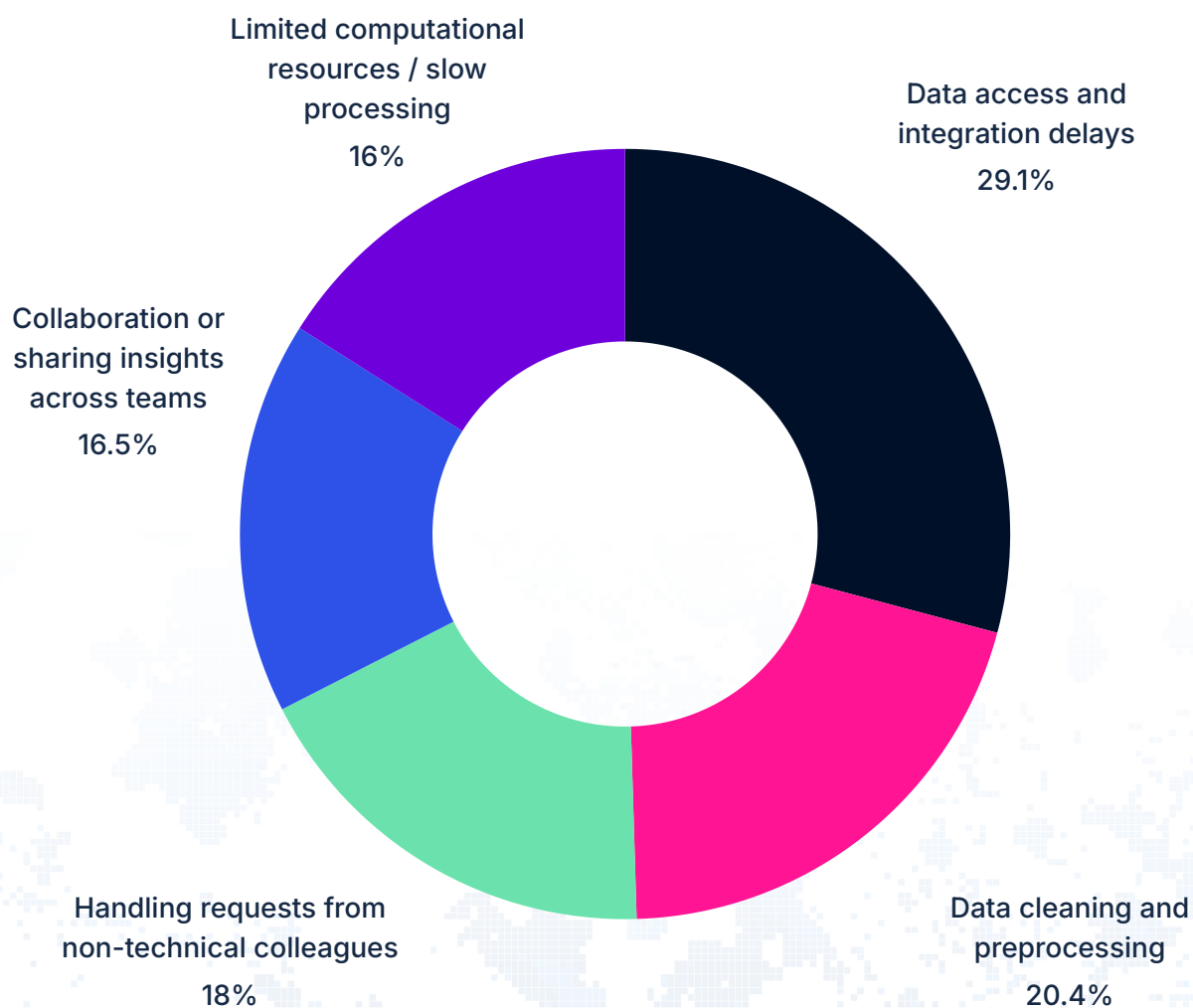
"Harmonization across different software"

Director of Data Science working in Natural Resources & Environment, USA

"Tooling is not all in one place, it's scattered. The offerings that do have a comprehensive suite of products are either expensive or not easily extendable. I don't want to be locked in by one vendor"

Data Engineer working in Geospatial Consulting, USA

Which aspect of your spatial analysis creates the biggest bottleneck for your organization?

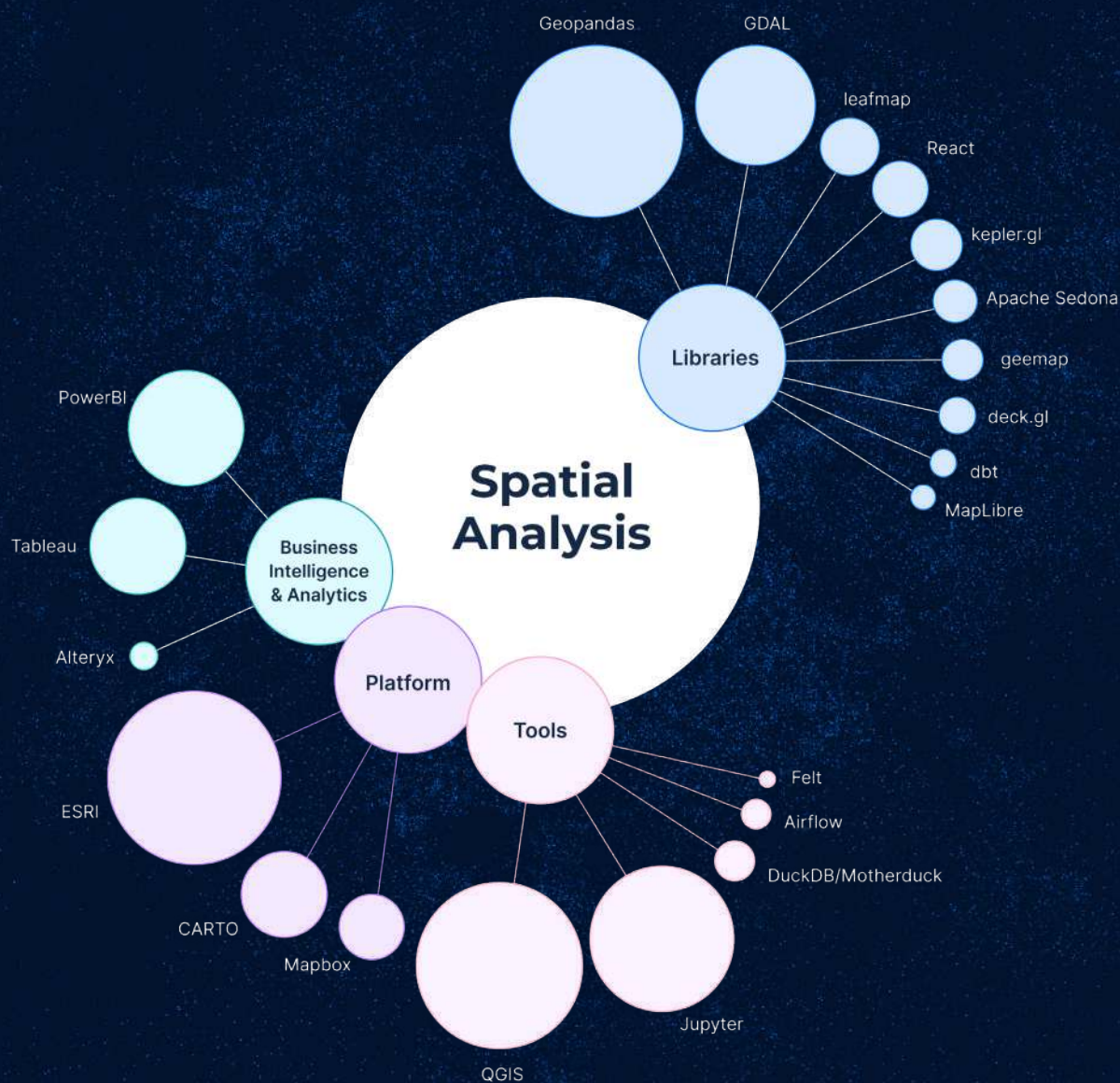


The new spatial data stack

This year's results show a spatial ecosystem that's bigger and broader than any previous survey.

QGIS, ESRI, GeoPandas, Jupyter, PowerBI, Tableau and GDAL are all popular choices - a mix of GIS staples, BI workhorses, and open-source libraries that reflect how spatial analysis has moved far beyond traditional GIS teams.

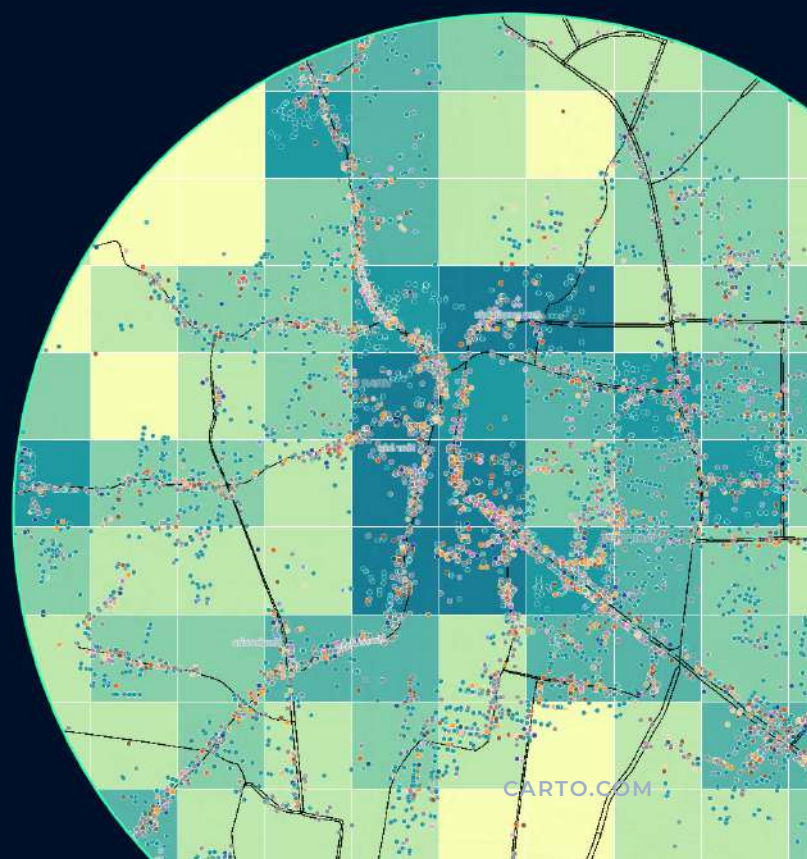
Rather than consolidating, organizations are expanding their toolkits to meet increasingly specialized needs: interactive web mapping with Mapbox and deck.gl, data engineering with Airflow, and data science in Jupyter, filling niche gaps.



*Nodes are sized by the number of respondents who use each tool

"The biggest challenge is fragmentation. The field is split between powerful but complex scientific tools (like those using Python/R) and user-friendly but often simplistic business intelligence (BI) platforms. This creates a gap where it's difficult to perform sophisticated, large-scale spatial analysis without deep technical expertise, slowing down decision-making and limiting its adoption across an organization."

Software Developer at a leading tech company

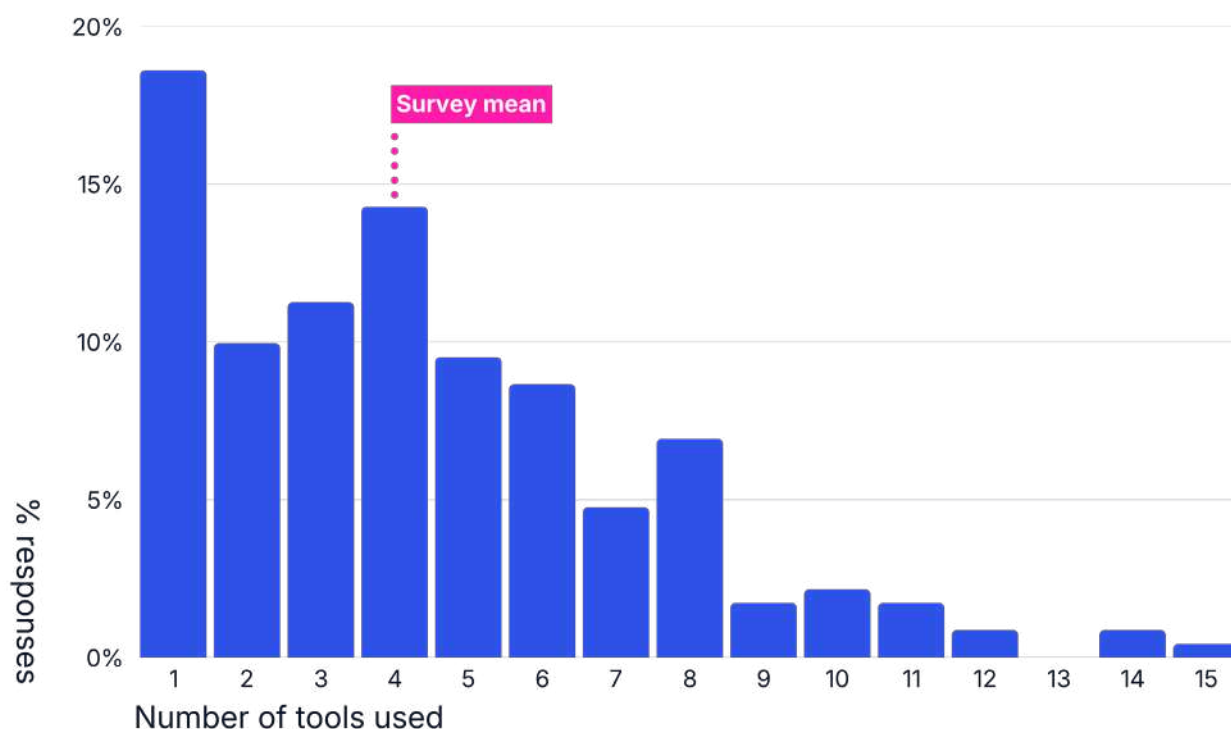


Your spatial stack only works when it works together

This year's survey makes one thing clear: spatial analysis **rarely happens in a single tool**. Fewer than 20% of respondents rely on just one platform, while the majority use between **3 and 8 tools to get work done** - and it's worth noting that the true number may be higher as only a limited selection of tools were given as options.

The spatial stack ranges from GIS tools like QGIS to BI tools, Python libraries, cloud data warehouses, and web-mapping frameworks. The result is powerful but fragile: insights depend on how well each component connects and shares data.

And this is where the real challenge emerges. A multi-tool stack only creates value when the pieces are actually interoperable - when data flows cleanly, duplicates are eliminated and governance stays intact. In 2026, **interoperability** isn't a "nice to have" - it's **crucial** and the only way an organization's spatial **stack** can deliver consistent, enterprise-grade insights.



Your 2026 spatial playbook:

Make your spatial stack work as one

Feeling the interoperability pain? Here's how to make 2026 the year your spatial stack works as one:

✓ **Step 1 • Create a single source of truth for your spatial data**

Fragmented inputs create fragmented insights. Centralize spatial storage and processing in your cloud warehouse or lakehouse, and connect tools to the data rather than shuttling data between tools.

✓ **Step 2 • Standardize shared models, governance, and workflows**

Different teams can use different tools - as long as they operate on the same logic. Define shared SQL models, naming conventions, governance rules, and pipeline patterns. Standardization lets every tool interoperate without forcing tool conformity.

✓ **Step 3 • Choose tools that embrace openness, APIs, and cloud-native integration**

Eliminate the pain of closed ecosystems, rigid extensions, and collaboration barriers with platforms like CARTO, which prioritize openness, accessibility & integrations via APIs, cloud-native connectors, AI providers, and LLMs, as well as MCP Servers. Build freely without lock-in.

✓ **Step 4 • Plug-in spatial analysis directly into enterprise systems**

Interoperability isn't just tool-to-tool - it's tool-to-business. Hook your spatial stack into BI tools, LLMs, AI agents, orchestration pipelines, geocoding services, and operational APIs so insights automatically flow into systems that make decisions.

✓ **Step 5 • Make collaboration effortless across teams**

Half of the integration bottlenecks aren't technical - they're human. Empower analysts, data scientists, GIS specialists, and business users to work from the same governed layers, share reproducible workflows, and collaborate without licensing blockers or brittle exports

The bottom line: Organizations in 2026 that invest in interoperability, not just individual tools, will move faster, automate more, and collaborate better.

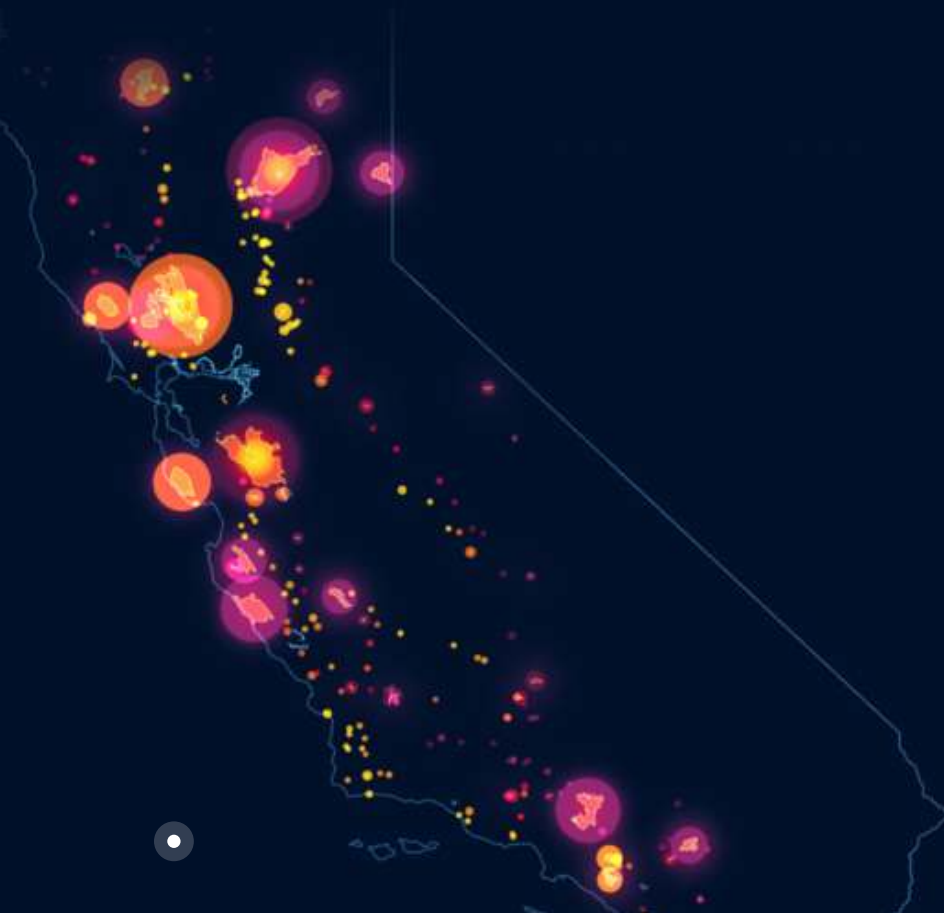


"All these tools are starting to come to maturity and build into systems... Before you had to duct tape a lot of solutions together and now you have these ***fully formed interoperable solutions.***"



Matt Forrest

Director of Customer Engineering & Product Led Growth at Wherobots





The democratization wave

Spatial Analytics is no longer confined to a single, specialized department; it has become a critical capability the entire organization relies on. While GIS professionals and data experts remain the foundation of this discipline, a surge of planners, product managers, engineers, strategists, and business leaders are now incorporating spatial thinking into their work. This creates a clear and urgent challenge: expertise remains concentrated in the hands of a few, while the need is democratization across organizations.

This widening gap between demand and supply is reflected in the persistent challenge organizations face when hiring spatial talent. This year's survey highlights this difficulty, with nearly half of respondents reporting struggles with hiring. This talent gap is compounded by continually expanding skill expectations for GIS roles, which now often include coding, cloud workflows, remote sensing, modeling, design, and deep domain knowledge. As a result, responsibilities that were once distributed across several specialized roles are now consolidated into a single job description.

With these versatile "spatial unicorns" in short supply, the only scalable path forward is to **empower** more people within the organization. The key is to amplify the impact of existing experts through modern tools and agentic capabilities, allowing their skills, context, and insights to be decentralized and scaled into repeatable workflows for all.



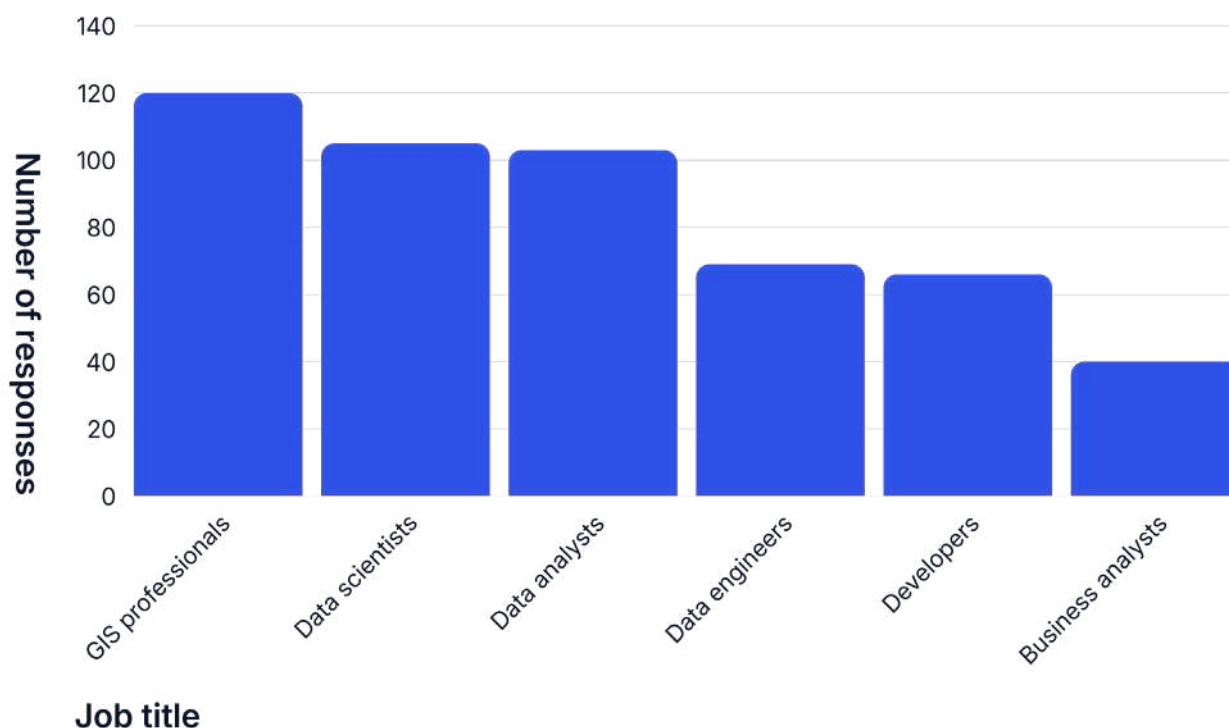
Who is “doing” spatial analysis?

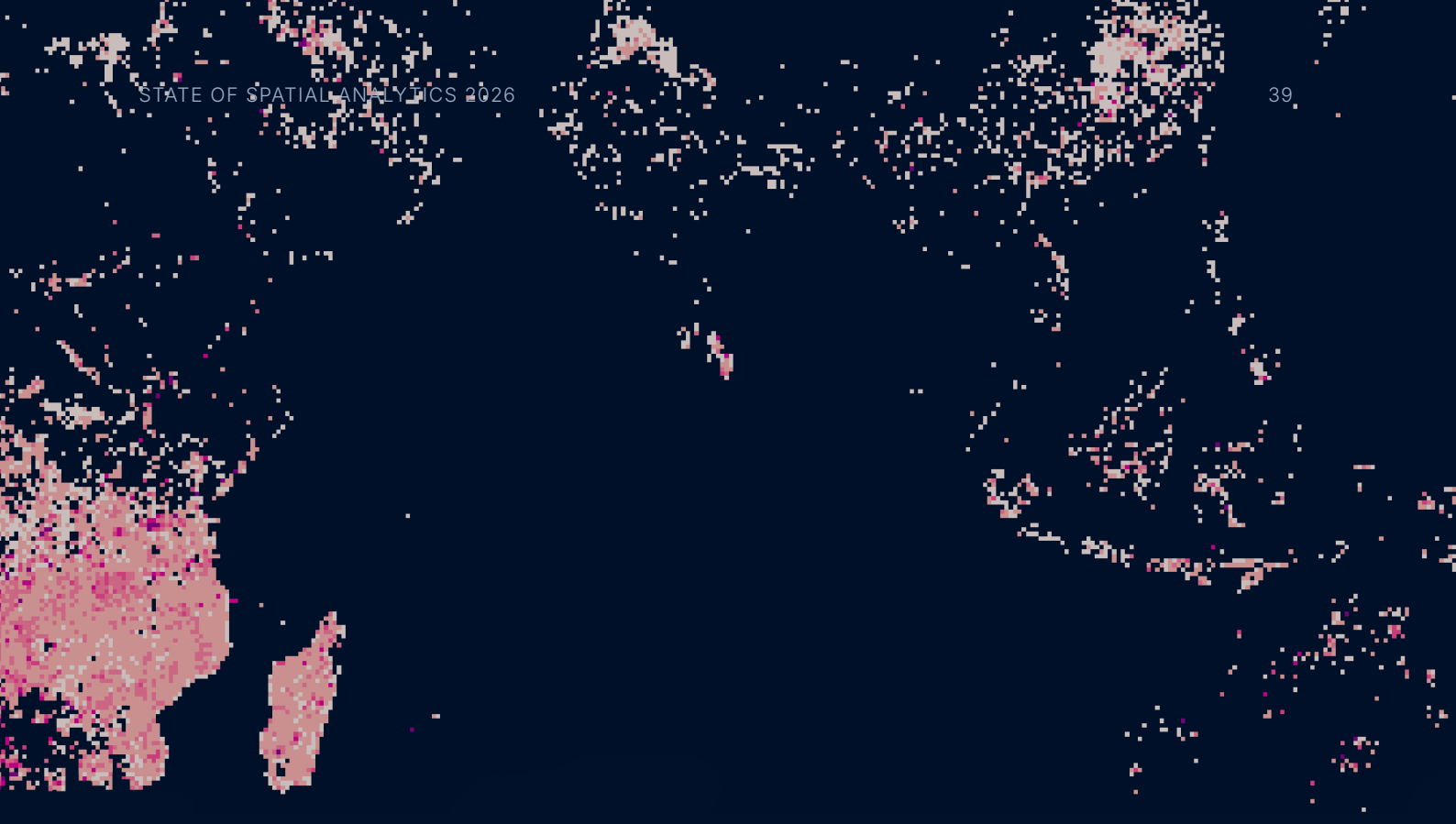
This year’s results make one thing unmistakably clear: **GIS experts remain the backbone** of spatial analysis, yet the audience is expanding fast.

GIS professionals, data scientists, data analysts, data engineers and developers still carry the technical load - but they’re no longer the only ones who need spatial insights to do their jobs.

The diversity of job titles of respondents in this year’s survey says it all. Climate analysts, CFOs, product managers, surveyors, humanitarian responders, ML engineers, real-estate strategists, planners, marketing leads, and even nurses all appear. The future of spatial isn’t just about empowering experts - it’s about equipping every team with the tools they need to think spatially, amplifying the knowledge & skills of GIS experts.

Who performs spatial analysis at your organization?





"Spatial Analytics is critical to our humanitarian work because it enables data-driven decisions in crisis settings, helping us allocate resources where they're needed most. ***It transforms complex geographic data into actionable insights***, improving the speed, precision, and impact of disaster preparedness and response."

GIS Officer at a leading global humanitarian initiative

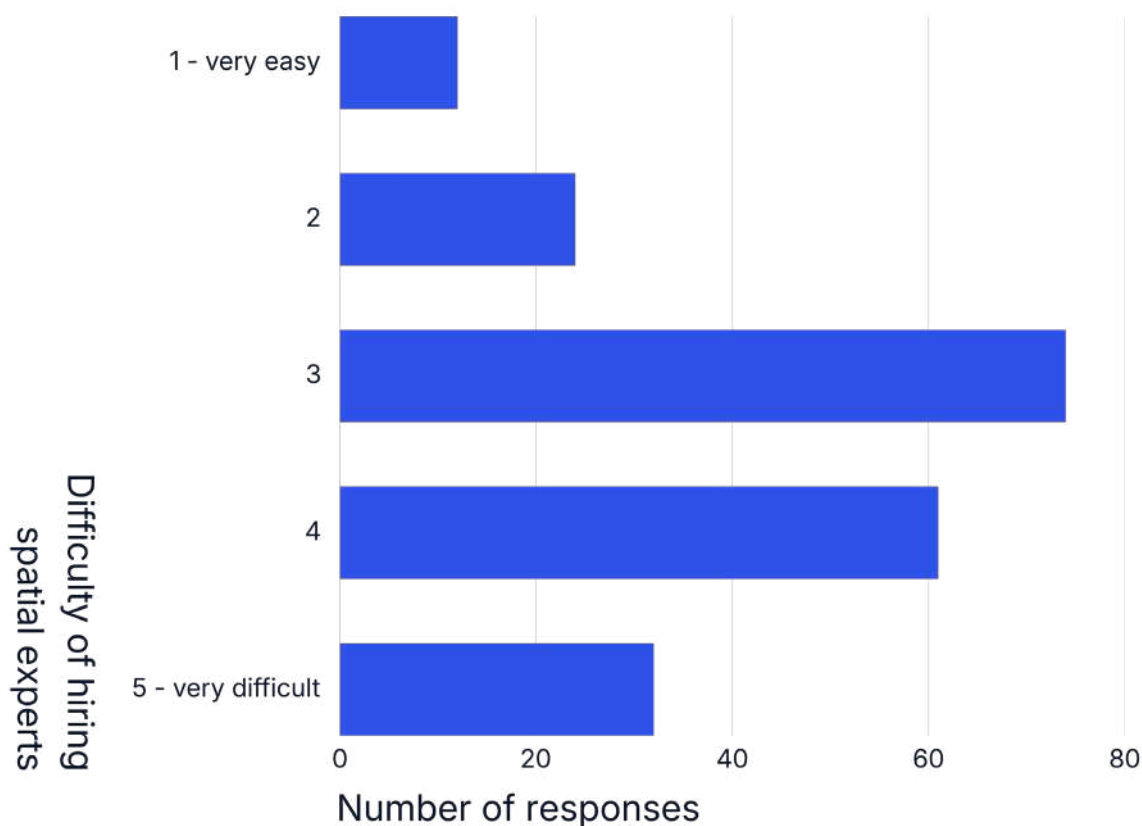
This word cloud illustrates the most common career titles from survey respondents. The size of each title is proportional to its frequency in the dataset.

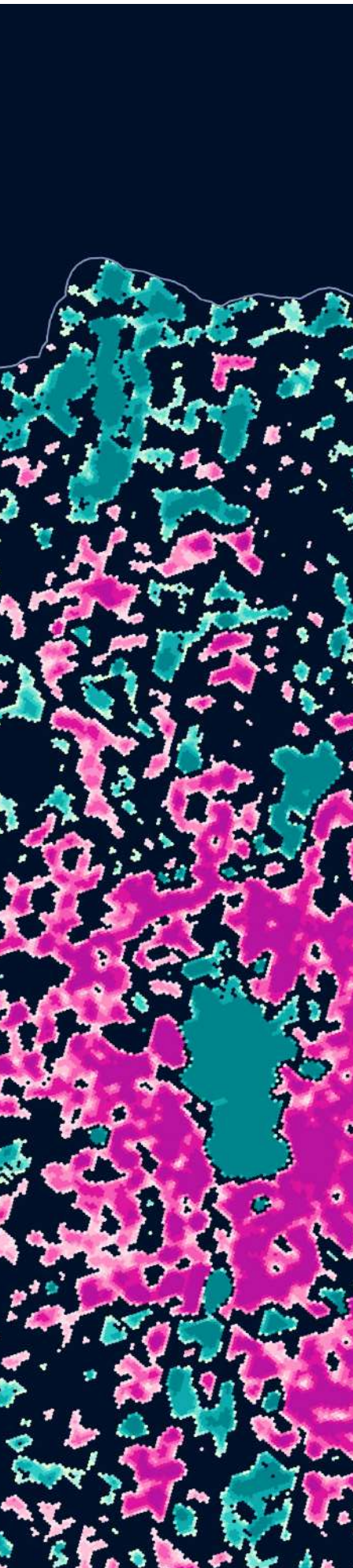


Hiring spatial experts remains a consistent challenge

In previous years, this survey has consistently found that organizations face major **challenges finding staff with spatial expertise** - and the problem isn't easing. This year, nearly half of respondents (46%) rated hiring spatially skilled talent as difficult or very difficult. There is no sign that the shortage of spatial experts is slowing.

As spatial analysis becomes a core part of data, engineering, and analytics teams, demand for skilled practitioners keeps rising. Yet the supply of people who can combine GIS fundamentals with modern data workflows simply isn't keeping pace. This talent bottleneck sits at the heart of the democratization story: organizations can't rely on hiring alone to expand spatial capability - they need to **empower** a much wider set of roles to contribute meaningfully to spatial work.





The demand for the spatial unicorn and why AI can help

Hiring challenges aren't just about scarcity - they're about **scope**. The responses to our survey show that modern GIS roles now demand an unusually wide skillset: coding, cloud processing, remote sensing, spatial analysis, cartography, data engineering - and even business storytelling. In 2026, the bar for a "spatial expert" has risen higher than ever.

Many teams describe candidates sitting at opposite ends of a spectrum - strong GIS but weak coding, or strong engineering but weak spatial fundamentals. And some organizations are still searching for the impossible: the all-singing, all-dancing spatial unicorn who can code, develop apps, design stunning maps, architect pipelines, and communicate to executives. This combination of skills is difficult enough to find in one expert, let alone across a whole team capable of servicing the - as we've already discussed - growing demands of a modern enterprise.

The real opportunity is to amplify the rare experts you do have - encoding their standards, domain knowledge, and organizational context into tools, workflows, and AI agents that scale their impact. Modern spatial stacks must **multiply expert capability**, not replace it.

Which skills are particularly difficult to find when recruiting?

"GIS skills along with environmental experience and good customer service"

Geospatial Consultant working in Geospatial Consulting, USA

"Geographers don't have expertise as data scientists and data scientists don't have GIS fundamentals"

Data Scientist, Ecuador

"GIS, MI/AL, modeling, spatial analytics, marketing and competitor knowledge"

Quantitative Analyst, USA

"Project management and Business Development skills alongside core geospatial tech"

Senior Earth Observation Consultant, Australia

"GIS specialists are just difficult to hire - there aren't many"

Engineering Manager, USA

Your 2026 spatial playbook:

Scaling expertise for enterprise-level impact

✓ **Step 1 • Built-in accessibility and integration**

Bring your spatial data out of isolation. Centralize all business data in a modern cloud data warehouse or lakehouse for easier integration. Run analysis directly so every team works from the same governed, scalable source of truth. Cloud-native processing removes friction, simplifies access, and makes spatial analysis available wherever your data already lives.

✓ **Step 2 • Empower teams with self-service spatial tools**

Put intuitive, AI-driven spatial capabilities in the hands of non-experts. Natural-language AI interfaces allow business users to ask spatial questions, explore patterns, and generate insights without relying on specialists for every request.

✓ **Step 3 • Use AI Agents to amplify your experts - not replace them**

Deploy AI Agents that inherit your experts' standards, logic, and domain knowledge to perform tasks, run tools, and support teams across the organization. This turns your scarce GIS specialists into force multipliers, scaling their impact without scaling headcount and ensuring high-quality spatial analysis everywhere it's needed.

The bottom line: Your experts can't be everywhere at once. But their expertise can. Deploy geospatial agents that embody your specialists' logic and business knowledge. This turns them into force multipliers, scaling their impact across the organization and ensuring high-quality spatial analysis is available everywhere.

"If I'm an analyst, this doesn't scare me.

This is fantastic. This gives me even more power to be able to do more analysis in a given day, leveraging that AI, fact-checking that AI, making sure that I can apply my expertise as an analyst."



Sean Wohltman,
Head of Cloud Geography at Google



05

Spatial data governance becomes mission-critical

Across this report, a clear theme emerges: the biggest challenge facing the spatial analytics community isn't just talent, tools, or AI adoption - it's **governance**.

Amid diverse tech stacks, massive data and increasing demand for Location Intelligence, organizations are struggling to define and enforce what 'good' governance looks like - and it matters now more than ever.

AI has made good governance more urgent. As workflows become increasingly automated, organizations cannot afford data silos, inconsistent definitions, or unvalidated datasets - all of which undermines the trust they place in the spatial insights they so desperately need.

53%

report data discovery & integration
as their top challenges when
working with spatial data.

47%

cite **data security & privacy**
as their biggest concern with
regards to AI.

36%

identify data discovery,
sourcing, and cleaning
as the most difficult part
of spatial analysis.

29.1%

cite data access &
integration delays as
their top bottleneck.

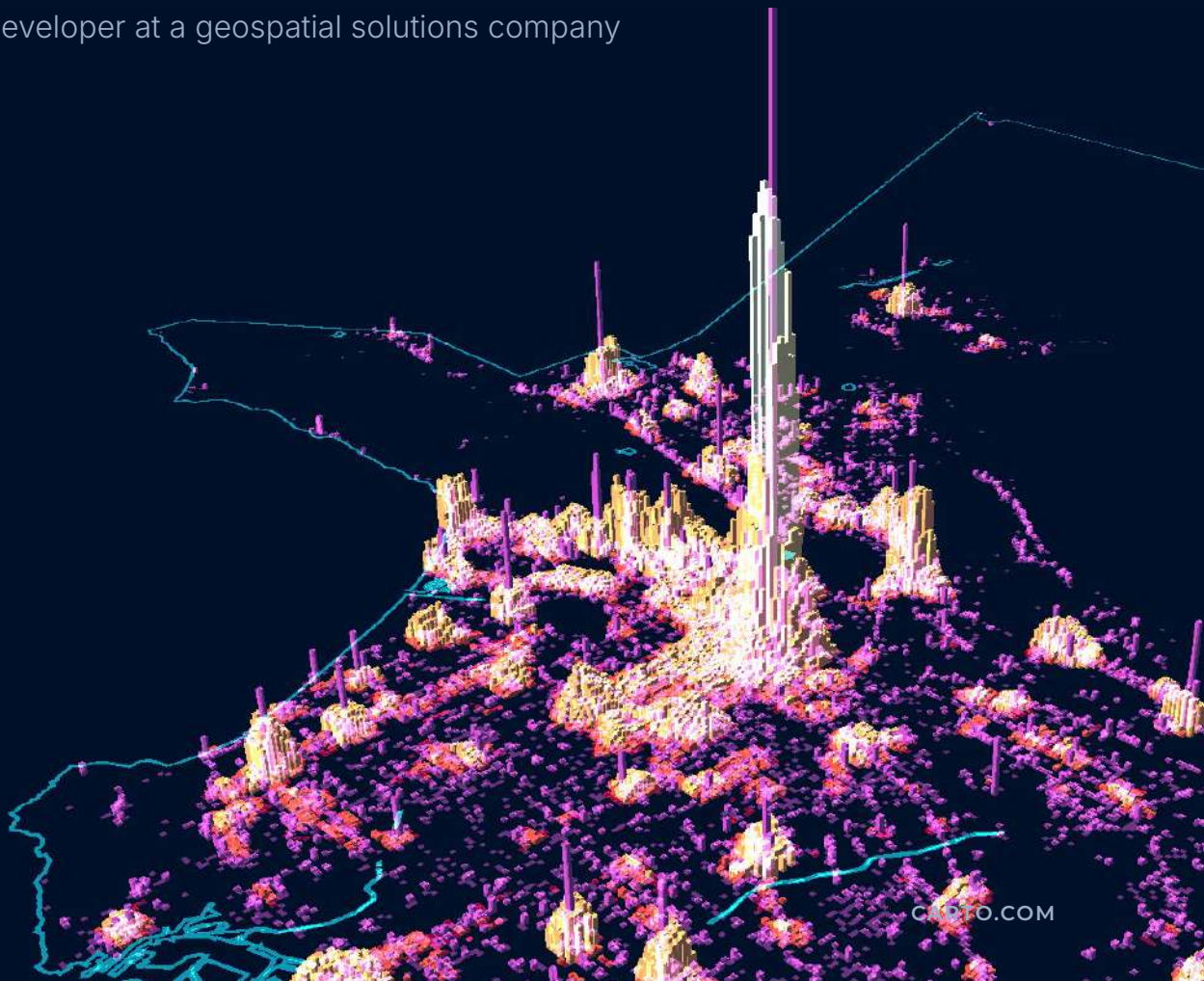
16.5%

additionally report
collaborating &
sharing insights as
the biggest problem.



"One of the biggest challenges currently facing the Spatial Analytics community is the integration and effective use of the massive volumes of spatial data being generated from diverse sources such as satellites, drones, IoT sensors, and mobile devices. While data availability has grown exponentially, ***many organizations still struggle with data standardization, interoperability, and quality control***, which are essential for accurate analysis."

Geospatial Developer at a geospatial solutions company



Key governance challenges

Data accessibility & integration

"Spatial data often remain siloed, and without clear governance and open data policies, it becomes difficult to build comprehensive, up-to-date national spatial databases that support decision-making"

Head of Geospatial Documentation & Information Services, Madagascar

Data quality & accuracy

"The biggest challenge is now what it always has been: data quality and veracity. Is the data true, valid, and complete?"

GIS Technical Coordinator, USA

Standardization & interoperability

"There is a lack of standardized data formats across different platforms and sources"

Chief Financial Officer, USA

Transparency & trust

"No fixed validation frameworks - results are always questionable due to the limitations of data and methods"

Product Manager (Geospatial Platforms), India

Collaboration & education

"We need to really drive forward the 'So what?'... it's one thing to have lots of data on a map, but another to create value-driving insight that's immediately accessible to non-technical users"

Insight & Data Manager, United Kingdom

Your 2026 spatial playbook:

Make your spatial stack work as one

Is data governance holding back from fully adopting AI? Here's where to start without compromising tech stack interoperability, data security, or analytics quality:

✓ **Step 1 • Define governance policies & frameworks**

Set clear policies for data quality, validation, and access. Standardize formats, coordinate reference systems, and naming conventions so every team works from the same playbook.

✓ **Step 2 • Centralize & catalog data, LLM, and frontier models**

Create a single source of truth using cloud-native warehouses or lakehouses. Capture metadata, lineage, LLM or frontier model methodology, and usage guidelines to make data discoverable, consistent, and reliable.

✓ **Step 3 • Automate quality & validation**

Build automated staging and development pipelines to check for accuracy, completeness, and bias. Include alerts and log issues so they are caught early on before they are rolled out to production.

✓ **Step 4 • Embed governance in tools & workflows**

Integrate rules into AI agents, spatial analytics platforms, and self-service tools. Ensure every transformation or analysis automatically follows your governance standards and is integrated with your required AI provider and LLM models with your business logic.

✓ **Step 5 • Make collaboration effortless across teams**

Train non-technical stakeholders in data literacy and AI trust. Foster a shared understanding of definitions, metrics, and thresholds across departments to reduce friction and maintain quality.

The bottom line: With AI raising the stakes, data governance is the critical foundation for trust and AI adoption. Embed automated governance directly into your spatial stack, making trusted insights the default, not the exception.

STATE OF SPATIAL ANALYTICS 2026

Conclusion



Amplifying Experts, Scaling Impact

Key takeaways

The Spatial Analytics community is at a pivotal moment. Across this year's survey, the underlying thread connecting all challenges is the need to define and operationalize what "good" looks like in a rapidly changing environment. Organizations that embed governance, empower teams with self-service tools, and amplify their experts with AI will not only survive but thrive as spatial analysis becomes core to decision-making. Those that rely on old silos, fragmented workflows, or ad-hoc practices risk falling behind.

Democratization is the new normal:

Scaling expertise via AI agents and self-service tools multiplies impact across the organization.

Data governance is critical:

Standardization, automation, and shared understanding reduce friction and enable actionable insights.

AI is a force multiplier, not a replacement:

Properly embedded, AI amplifies experts and operationalizes organizational knowledge.

Cloud-native, centralized, and accessible data is a prerequisite:

Organizations with accessible, governed spatial data are consistently more effective and agile.

Spatial talent is as scarce as ever:

Spatial unicorns are rare, and demand continues to outstrip supply.





The road ahead

That concludes the State of Spatial Analytics for 2026! But... What happens next? Here are our top predictions for what the geospatial world will look like when we report back in 2028.

- 1. AI as a differentiator:** Early adopters of AI will see productivity and collaboration soar, while organizations slow to embrace these tools risk falling behind.
- 2. The GIS expert evolves:** GIS specialists will shift from building one-and-done maps and dashboards to curating geospatial reasoning experiences that scale across the enterprise.
- 3. Governance goes autonomous:** Data governance will become less manual and more embedded, with policies, validation, and quality checks built into workflows by default.
- 4. Cloud-native as standard:** Legacy stacks will no longer support the scale modern enterprises demand; cloud-native architectures will dominate spatial analytics.
- 5. Interoperability expected, not optional:** As the sector diversifies, so too will the tooling ecosystem - seamless integration across platforms will be table stakes.



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